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SUN YAT-SEN UNIVERSITY

代谢功能障碍相关脂肪性肝病与 四种常见疾病的风险关联



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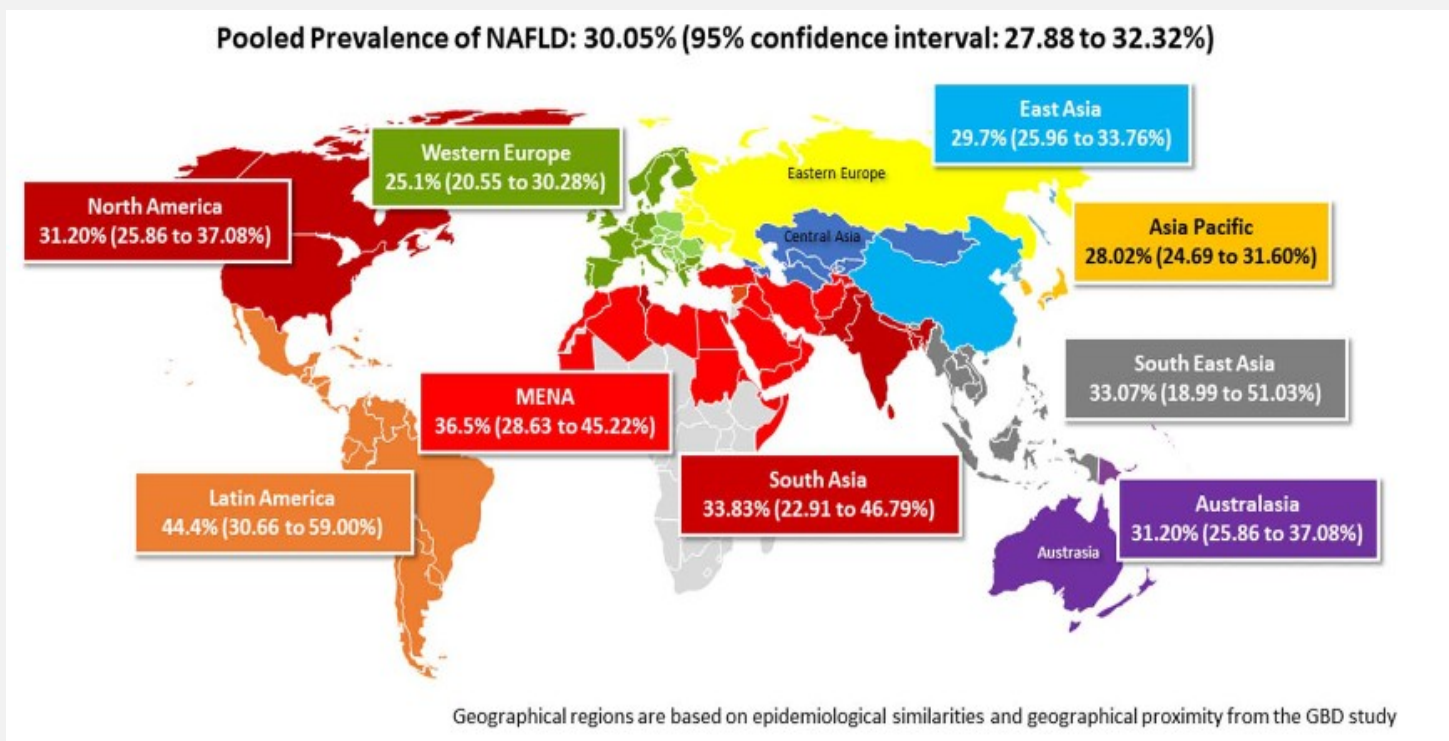
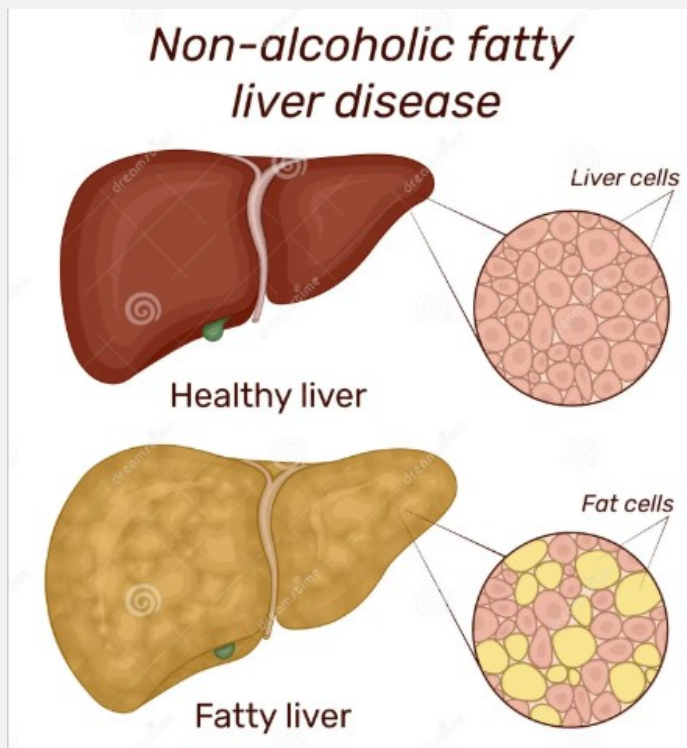
Part.01

研究背景



1.1 疾病更名过程

非酒精性脂肪性肝病 (Non-alcoholic fatty liver disease ; NAFLD)



NAFLD 是目前全球范围内最常见的慢性肝病，预计影响全球 38% 的人口。

NAFLD 及其引起的非酒精性脂肪性肝炎（NASH）是肝细胞癌（HCC）的病因，也是肝移植的主要适应症。



1.1 疾病更名过程

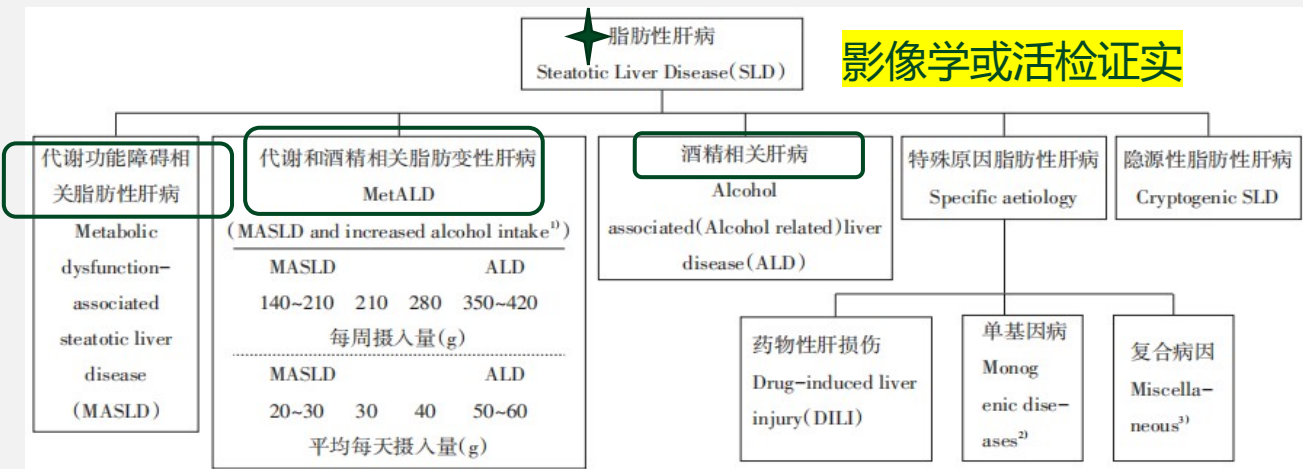
非酒精性脂肪性肝病 (NAFLD) 1980

排他性

代谢功能障碍相关脂肪性肝病 (MASLD) 2023

Metabolic dysfunction-associated steatotic liver disease 肯定性

NAFLD 的诊断需要有弥漫性肝细胞脂肪变的影像学或组织学证据，并且要排除乙醇（酒精）滥用等可以导致肝脂肪变的其他病因。



MASLD/MetALD/ALD 诊断标准

因素 疾病	肝脂肪变性	至少存在 5 种心血管代谢危险因素 (CMRF) 中的 1 种	酒精摄入量大小
MASLD	√	√	女性少于 140 g/周, 男性少于 210 g/周
MetALD	√	√	女性: 140-350 克/周; 男性: 210-420 克/周
ALD	√	×	女性超过 350 克/周, 男性超过 420 克/周

1.1 疾病更名过程

心血管代谢危险因素（CMRF）成人标准	
(1)	BMI $\geq$ 25kg /m ² （亚洲人为 23kg /m ² ）或腰围 $\geq$ 94 cm（男性），80 cm（女性）或根据种族特点调整
(2)	空腹血清葡萄糖 $\geq$ 5.6 mmol/L（100 mg/dL）或餐后 2 小时血糖 $\geq$ 7.8 mmol/L（ $\geq$ 140 mg/dL）或糖化血红蛋白 $\geq$ 5.7%（39 mmol/L）或患有 2 型糖尿病或正在接受 2 型糖尿病治疗
(3)	血压 $\geq$ 130/85 mmHg 或特异性抗高血压药物治疗
(4)	血浆甘油三酯 $\geq$ 1.70 mmol/L（150 mg/dL）或降脂治疗
(5)	血浆高密度脂蛋白 - 胆固醇 $\leq$ 1.0mmol/L[40 mg/dL]（男性）， $\leq$ 1.3 mmol/L[50 mg/dL]（女性）或降脂治疗



1.2 MASLD、MetALD 和 ALD 与四种常见疾病结局的关联研究

作者	队列	研究结论
Jingli Gao 等	中国开滦	MASLD 与慢性肾脏疾病（CKD）的风险增加有关
Yewan Park 等	韩国	MASLD、MetALD 与更高的肝癌及肝外肿瘤，尤其是胃肠道肿瘤的发生风险相关
Joon Ho Moon 等	韩国	MASLD 增加了心血管疾病的风险
Hyeok-Hee Lee 等	韩国	MASLD 增加了心血管疾病的风险

研究现状：

- 这些研究是基于脂肪肝指数（FLI）或者超声来定义脂肪肝变性，与肝活检和磁共振成像相比，其准确性及灵敏度不足。
- MASLD 与肝病结局之间关联分析的队列研究目前尚未找到



1.2 MASLD、MetALD 和 ALD 与四种常见疾病结局的因果关联研究

作者	研究方法	研究结论
Lijun Zhao 等	双样本孟德尔随机化	NAFLD 与心脑血管事件（冠心病、心力衰竭、心肌梗死、心绞痛和中风）之间存在因果关系
Jiarong Xie 等	双样本孟德尔随机化	NAFLD 与女性乳腺癌、宫颈癌、喉癌、白血病、肺癌和前列腺癌的风险增加有因果关联

研究现状：

- 尚未有研究基于 MASLD GWAS 汇总数据探索 MASLD 与四种肝内外疾病结局的因果关联



1.4 UKB 队列简介



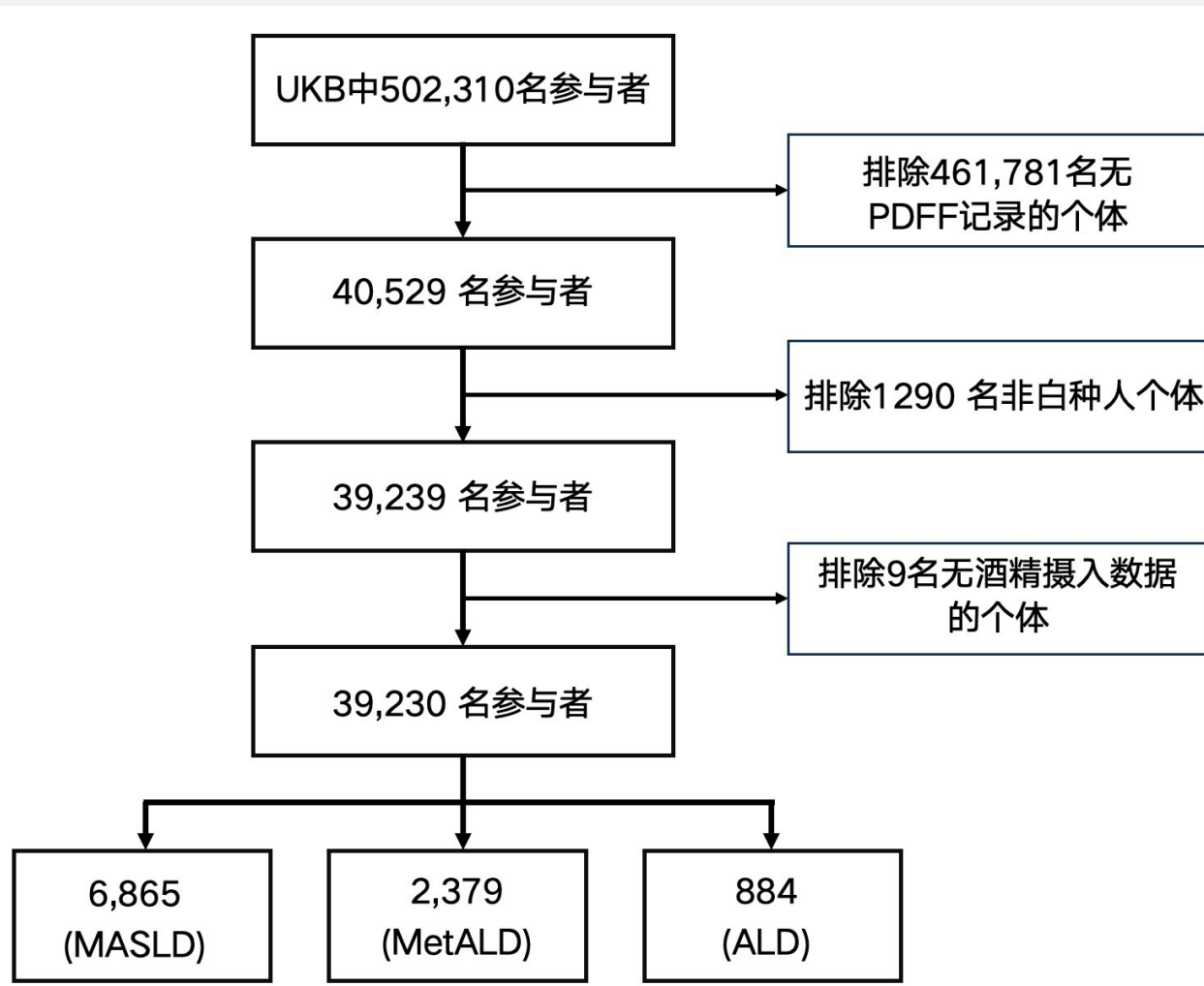
大规模

前瞻性研究

2006-2010 年期间

50 多万人 40-69 岁成年人

提供基因型和表型数据





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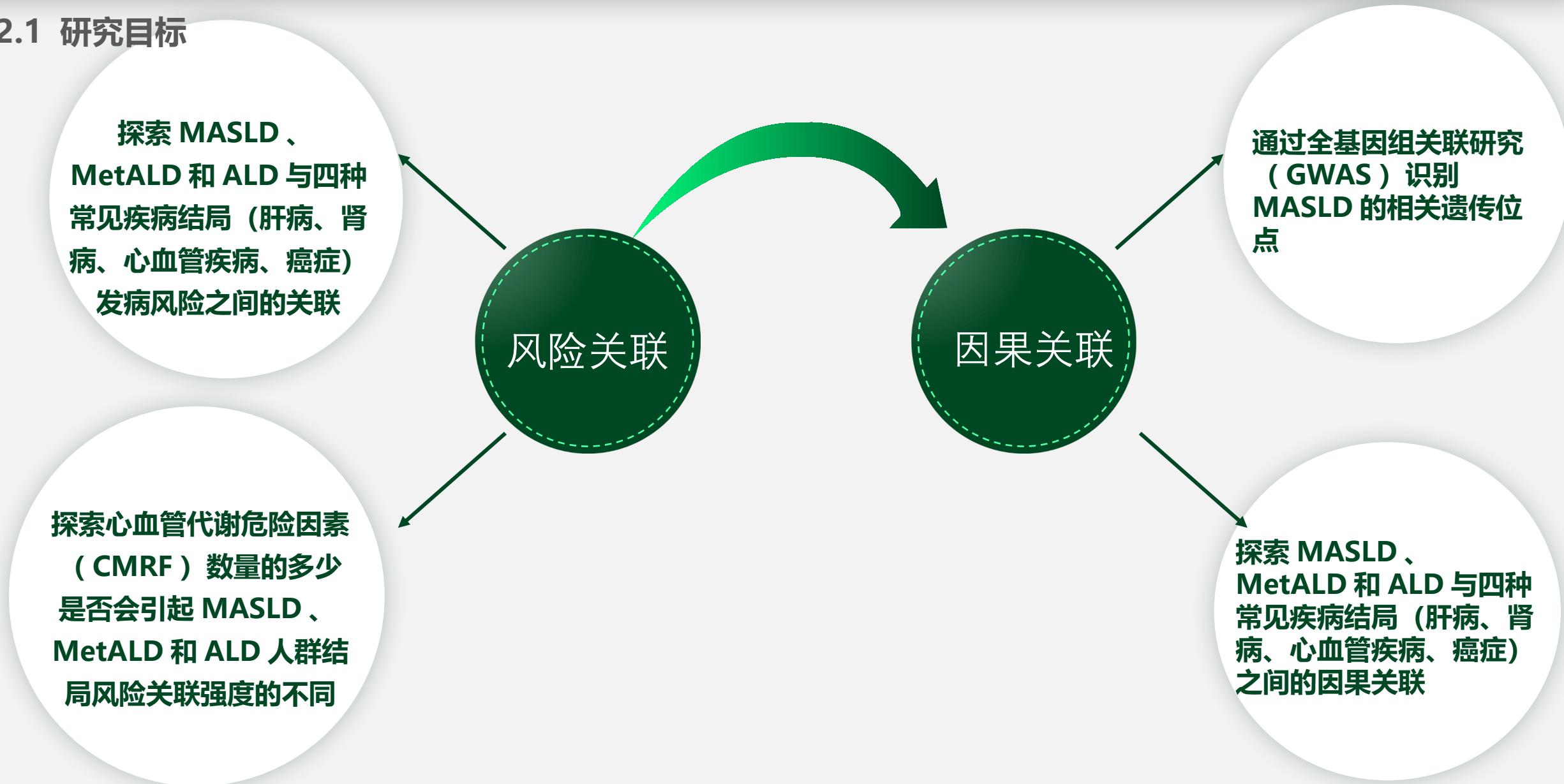
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Part.02

研究目标



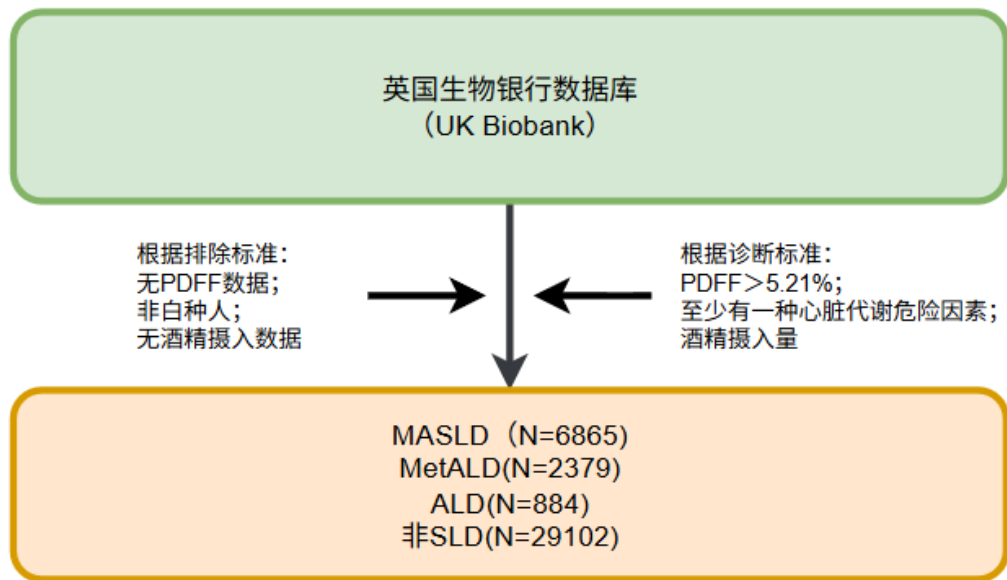
2.1 研究目标



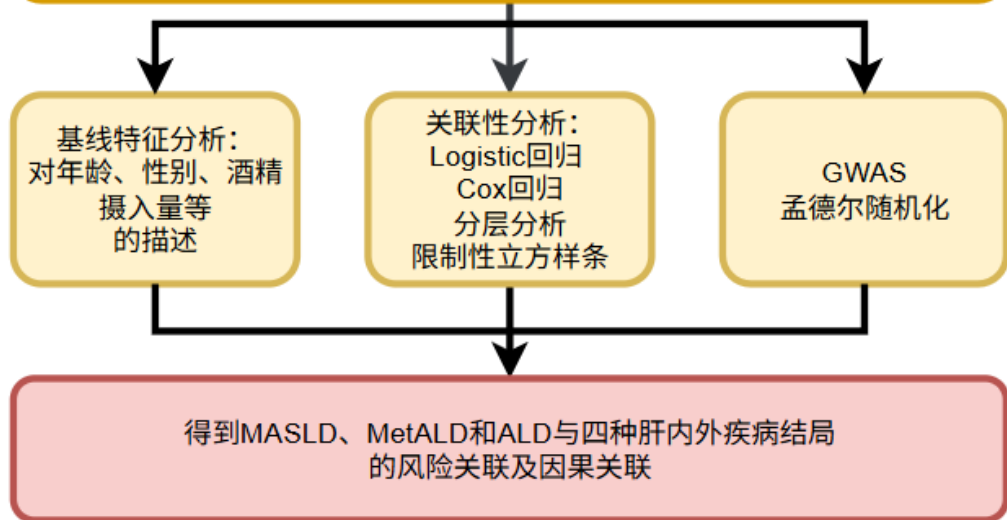


2.2 技术路线图

研究人群筛选:



数据分析:





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研究工作进展



暴露： (MASLD/MetALD/ALD)

- 磁共振成像 -- 质子密度脂肪分数
PDFF>5.21%
- 至少一种心脏代谢危险因素 (CMRF)
- 酒精摄入量大小

结局变量：

疾病结局	ICD-10
肝病	B18.x, C22.x, K71.x–K75.x, K76.1–K76.9, K77.X, Z94.4
肾病	I12.0, I13.1, N03.2–N03.3, N03.5–N03.7, N05.2–N05.7, N18.x–N19.x, N25.0, Z49.x, Z94.0, Z99.2
心血管疾病	I09.9, I11.0, I13.0, I13.2, I21.x–I22.x, I25.2, I25.5, I42.0, I42.5–I42.9, I43.x, I50.x
癌症	C00.x–C97.x



表 1. 参与者的基线特征

Characteristics	All (n = 39230)	Other (n = 29102)	MASLD (n = 6865)	MetALD (n = 2379)	ALD (n = 884)
Age at baseline, y, mean ($\pm$ SD)	55.12 $\pm$ 7.51	55.05 $\pm$ 7.58	55.58 $\pm$ 7.39	54.90 $\pm$ 7.09	54.35 $\pm$ 7.04
Gender, n (%)					
Male	18733(47.75)	12606(43.32)	3876(56.46)	1551(65.20)	700(79.19)
Female	20497(52.25)	16496(56.68)	2989(43.54)	828(34.80)	184(20.81)
Smoking status					
Prefer not to answer	19(0.05)	13(0.04)	4(0.06)	1(0.04)	1(0.11)
Never	23319(59.44)	17784(61.11)	4207(61.28)	1018(42.79)	310(35.07)
Previous	14522(37.02)	10342(35.54)	2442(35.57)	1228(51.62)	510(57.69)
Current	1370(3.49)	963(3.31)	212(3.09)	132(5.55)	63(7.13)
Alcohol frequency					
Daily or almost daily	9086(23.16)	6565(22.56)	801(11.67)	1073(45.10)	647(73.19)
Three or four times a week	11192(28.53)	8547(29.37)	1496(21.79)	950(39.93)	199(22.51)
Once or twice a week	10125(25.81)	7608(26.14)	2157(31.42)	326(13.70)	34(3.85)
One to three times a month	4201(10.71)	3058(10.51)	1116(16.26)	24(1.01)	3(0.34)
Special occasions only	3001(7.65)	2142(7.36)	852(12.41)	6(0.25)	1(0.11)
Never	1625(4.14)	1182(4.06)	443(6.45)	0(0.00)	0(0.00)
Overall health rating					
Do not know	2(0.01)	1(0.00)	1(0.01)	0(0.00)	0(0.00)
Excellent	7499(19.12)	6328(21.74)	759(11.06)	310(13.03)	102(11.54)
Good	24905(63.48)	18670(64.15)	4212(61.35)	1505(63.26)	518(58.60)
Fair	6099(15.55)	3712(12.76)	1657(24.14)	491(20.64)	239(27.04)
Poor	725(1.85)	391(1.34)	236(3.44)	73(3.07)	25(2.83)
BMI category, mean ($\pm$ SD)	26.41 $\pm$ 4.28	25.27 $\pm$ 3.60	29.83 $\pm$ 4.50	29.46 $\pm$ 4.10	29.30 $\pm$ 4.14
MET per week	2855.15 $\pm$ 2511.00	2972.65 $\pm$ 2526.00	2498.26 $\pm$ 2455.00	2577.30 $\pm$ 2364.00	2489.25 $\pm$ 2482.00
Townsend index	2.36 $\pm$ 1.83	2.36 $\pm$ 1.84	2.40 $\pm$ 1.84	2.31 $\pm$ 1.77	2.33 $\pm$ 1.81
Albumin	45.49 $\pm$ 2.54	45.41 $\pm$ 2.53	45.59 $\pm$ 2.55	45.90 $\pm$ 2.47	46.05 $\pm$ 2.56
Alkaline phosphatase	80.03 $\pm$ 23.26	78.81 $\pm$ 22.83	85.17 $\pm$ 25.26	80.46 $\pm$ 20.65	79.41 $\pm$ 22.44
Alanine aminotransferase	22.82 $\pm$ 13.47	20.52 $\pm$ 11.06	29.01 $\pm$ 16.94	29.79 $\pm$ 16.95	31.94 $\pm$ 18.62
Aspartate aminotransferase	25.77 $\pm$ 10.14	24.96 $\pm$ 8.30	27.73 $\pm$ 15.02	28.38 $\pm$ 11.00	30.20 $\pm$ 11.50
Gamma glutamyltransferase	33.35 $\pm$ 33.11	29.72 $\pm$ 28.77	40.00 $\pm$ 36.04	46.91 $\pm$ 38.16	65.23 $\pm$ 71.49
Glucose	5.00 $\pm$ 0.96	4.95 $\pm$ 0.88	5.16 $\pm$ 1.23	5.11 $\pm$ 0.90	5.16 $\pm$ 1.03
HbA1c	35.07 $\pm$ 5.08	34.67 $\pm$ 4.60	36.57 $\pm$ 6.49	35.58 $\pm$ 5.05	35.15 $\pm$ 5.37
HDL cholesterol	1.49 $\pm$ 0.38	1.55 $\pm$ 0.38	1.28 $\pm$ 0.30	1.39 $\pm$ 0.34	1.46 $\pm$ 0.39
Phosphate	1.17 $\pm$ 0.16	1.17 $\pm$ 0.16	1.15 $\pm$ 0.17	1.15 $\pm$ 0.16	1.13 $\pm$ 0.17
Triglycerides	1.64 $\pm$ 0.95	1.47 $\pm$ 0.80	2.16 $\pm$ 1.12	2.06 $\pm$ 1.13	2.11 $\pm$ 1.22
PDFF	4.92 $\pm$ 4.93	2.73 $\pm$ 0.98	11.11 $\pm$ 6.12	11.05 $\pm$ 6.08	12.11 $\pm$ 6.96



图 2. 参与者患四种常见疾病的比例

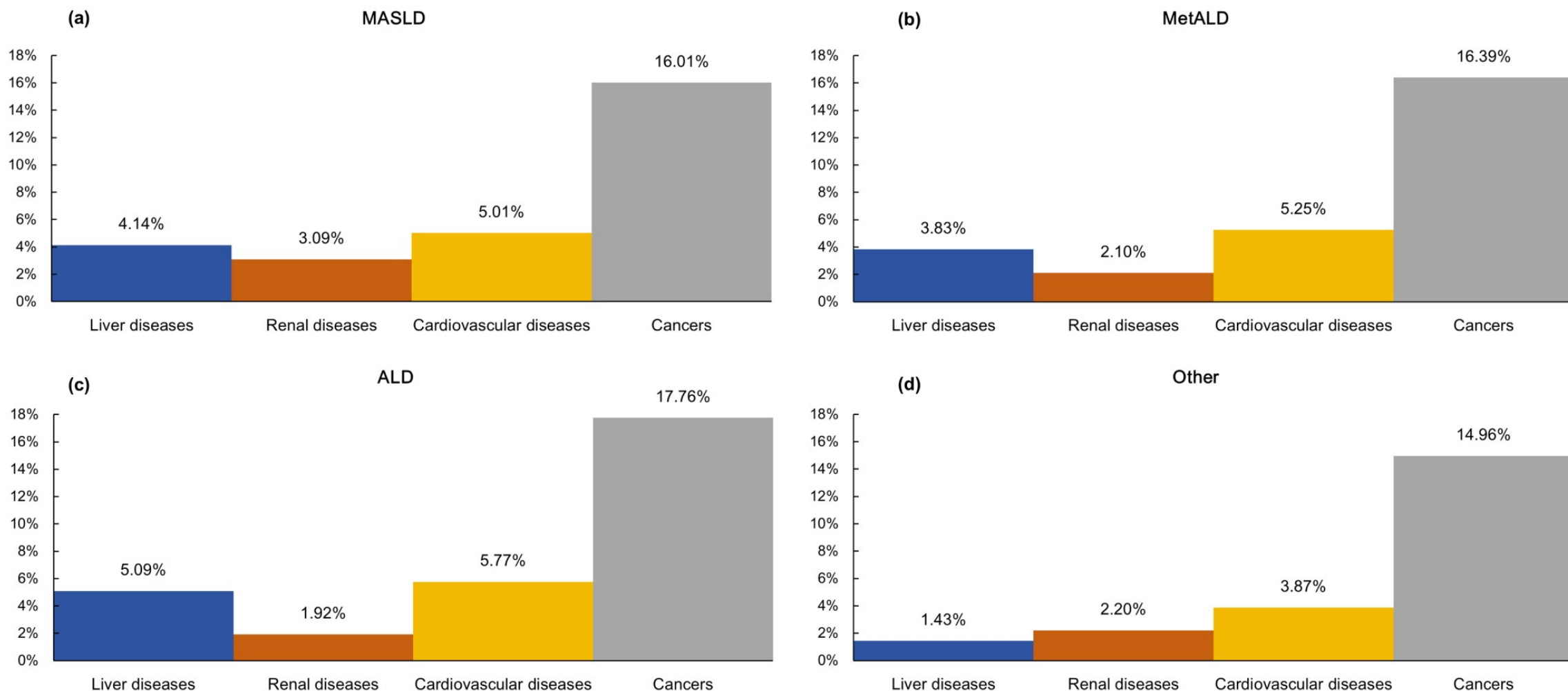




表 2. logistic 回归分析 SLD （ MASLD/MetALD/ALD ） 与四种疾病之间的关系

Outcomes	Diagnoses	Unadjusted		Adjusted	
		OR (95%CI)	P value	OR (95%CI)	P value
Liver diseases	MASLD	2.97(2.55,3.46)	<.0001	3.98(2.55,3.48)	<.0001
	MetALD	1.65(1.47,1.86)	<.0001	2.81(2.22,3.54)	<.0001
	ALD	1.55(1.39,1.72)	<.0001	3.87(2.81,5.33)	<.0001
Renal diseases	MASLD	1.42(1.21,1.66)	<.0001	1.34(1.14,1.57)	3.0×10 ⁻⁴
	MetALD	0.98(0.85,1.13)	0.76	0.97(0.72,1.30)	0.82
	ALD	0.96(0.81,1.12)	0.58	0.91(0.56,1.49)	0.71
Cardiovascular diseases	MASLD	1.31(1.16,1.48)	<.0001	1.15(1.01,1.30)	0.04
	MetALD	1.17(1.07,1.29)	9.0×10 ⁻⁴	1.17(0.97,1.42)	0.11
	ALD	1.15(1.04,1.27)	4.4×10 ⁻³	1.19(0.89,1.60)	0.24
Cancers	MASLD	1.08(1.01,1.16)	0.03	1.04(0.97,1.12)	0.31
	MetALD	1.06(1.00,1.12)	0.06	1.12(1.00,1.26)	0.06
注：调整的协变量为：年龄和性别。		1.07(1.01,1.14)	0.02	1.27(1.06,1.52)	9.0×10 ⁻³



表 3. COX 回归分析 SLD （ MASLD/MetALD/ALD ） 与四种疾病之间的关系图 1.SLD 与四种疾病结局的 Kaplan-Meier 生存曲线

Outcomes	Diagnoses	Unadjusted		Adjusted	
		HR (95%CI)	P value	HR (95%CI)	P value
Liver diseases	MASLD	3.04(2.60,3.56)	<.0001	3.05(2.60,3.57)	<.0001
	MetALD	2.69(2.12,3.42)	<.0001	2.77(2.17,3.52)	<.0001
	ALD	3.99(2.92,5.45)	<.0001	4.21(3.06,5.78)	<.0001
Renal diseases	MASLD	1.40(1.20,1.64)	<.0001	1.32 (1.13,1.55)	5.0×10 ⁻⁴
	MetALD	0.94(0.70,1.27)	0.70	0.96(0.71,1.29)	0.78
	ALD	0.85(0.52,1.40)	0.53	0.90(0.55,1.48)	0.68
Cardiovascular diseases	MASLD	1.23(1.08,1.40)	1.6×10 ⁻³	1.08(0.95,1.23)	0.26
	MetALD	1.33(1.10,1.62)	4.0×10 ⁻³	1.13(0.93,1.37)	0.22
	ALD	1.47(1.09,1.97)	0.01	1.16(0.87,1.57)	0.32
Cancers	MASLD	1.07(1.00,1.16)	0.06	1.02(0.95,1.10)	0.58
	MetALD	1.09(0.97,1.23)	0.14	1.07(0.95,1.20)	0.24
	ALD	1.36(1.15,1.60)	3.0×10 ⁻⁴	1.34(1.14,1.58)	5.0×10 ⁻⁴

注：调整的协变量为：年龄和性别。

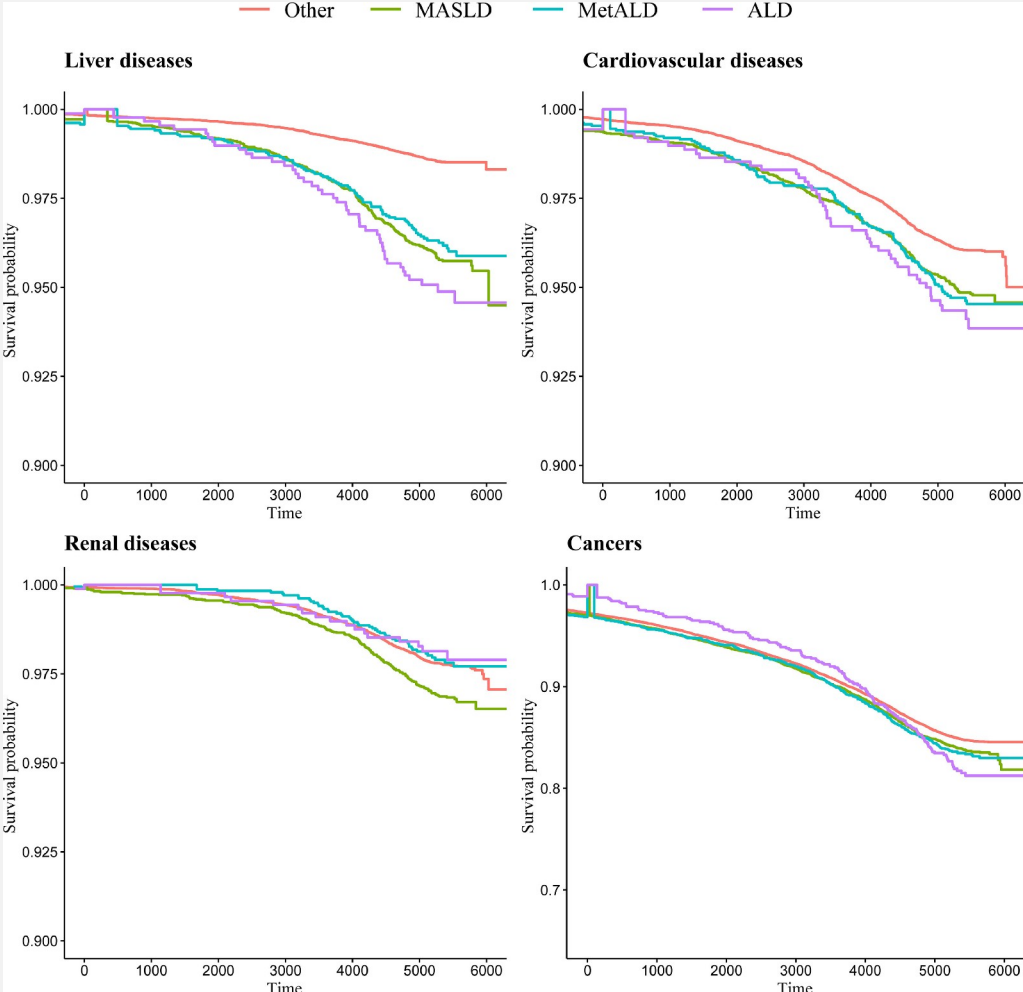
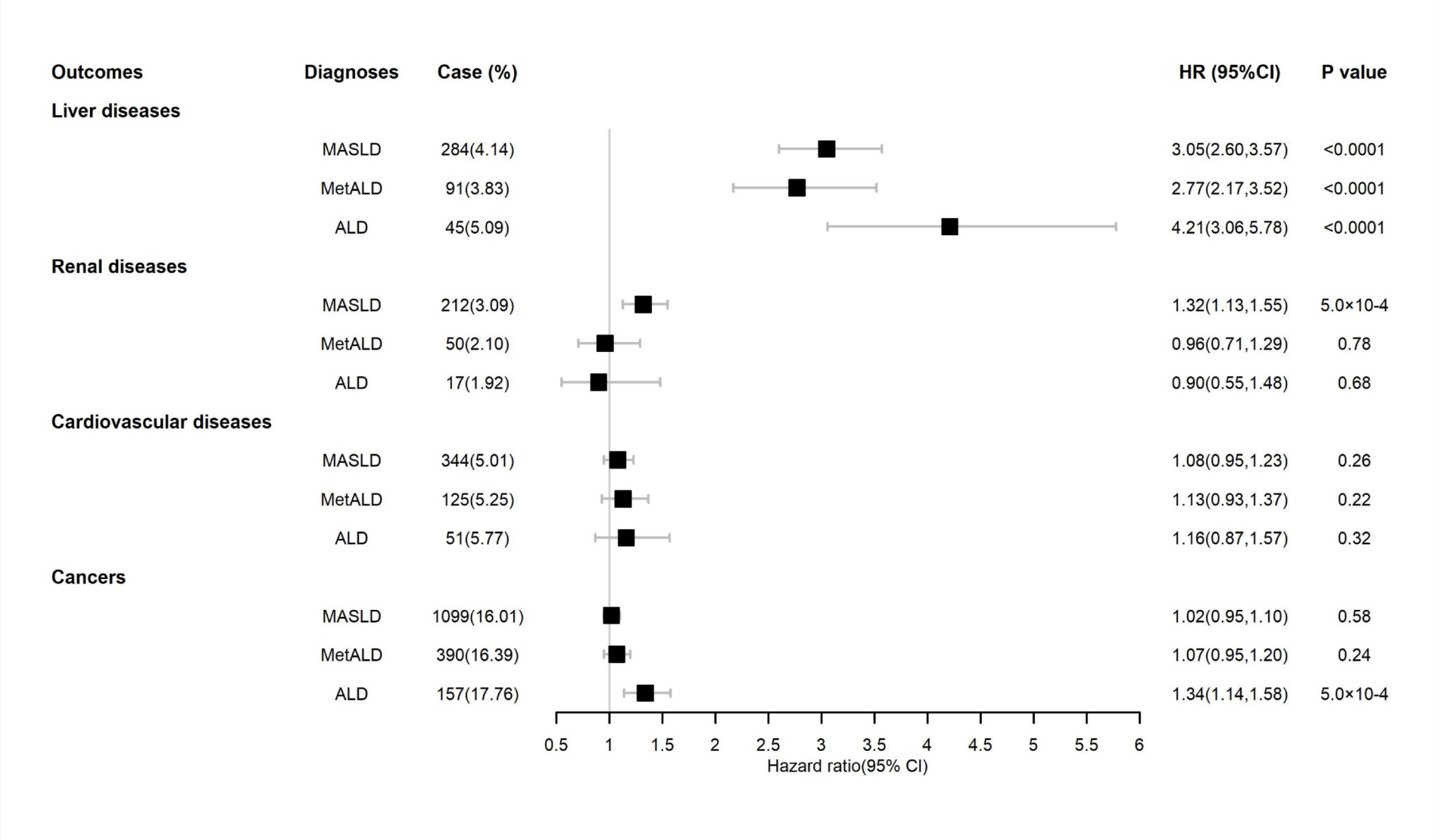


图 1.MASLD、MetALD 和 ALD 与四种肝内和肝外疾病的关联





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后续研究计划



已获得英国生物银行（UK Biobank, UKB）数据的使用授权，并可合法合规地使用相关数据开展研究。可访问 UKB 提供的完整数据，包括高质量的影像学资料、表型信息、基因组学数据以及长期随访的疾病结局记录等。

数据获取

01

02

研究基础

已成功从 UKB 中提取研究需要的变量数据，并已完成前期的数据清洗编码处理，完成关联分析

后续研究计划

03

2023.12~2024.5

已完成：队列数据整理、数据清洗、统计分析

2024.6~2024.12

已完成：关联分析、第一篇文章发表

2025.01~2025.06

未完成：敏感性分析、GWAS 和孟德尔随机化分析

2025.07~2026.5

未完成：作图制表、毕业论文撰写与修改



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